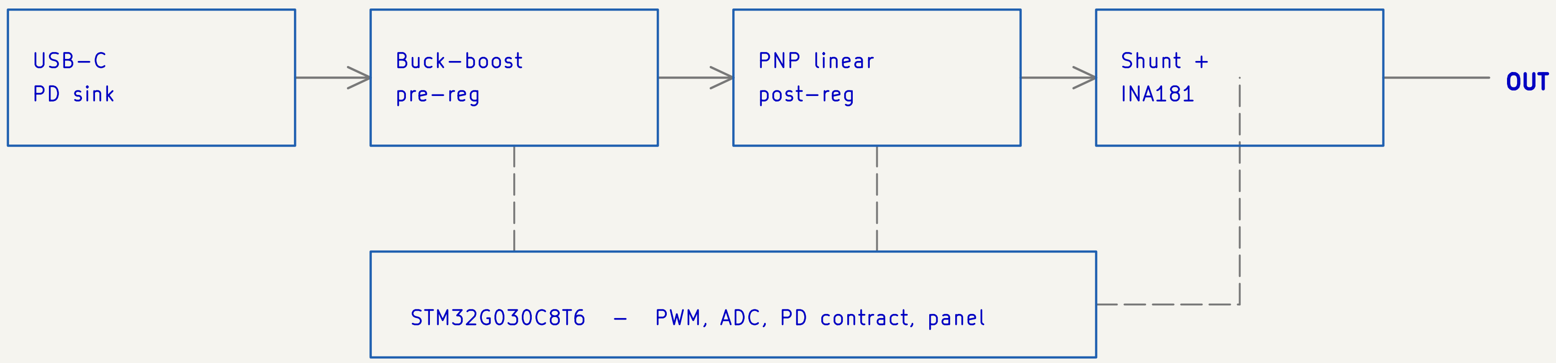
USB-C PD BENCH POWER SUPPLY

0 – 36 V / 0 – 3 A adjustable bench supply from a USB-C PD source

SPECIFICATION

Input	USB-C PD sink, 5 / 9 / 12 / 15 / 20 V contracts
Pre-regulator	4-switch synchronous buck-boost, 250 kHz
Post-regulator	PNP Darlington linear stage, tracking headroom
Output	0 – 36 V, 0 – 3 A
Set resolution	10 mV / 10 mA, PWM reference + 2-pole RC
Readback	voltage and current, 12-bit ADC
Protection	hardware over-current trip, latching
Modes	constant voltage, constant current, battery emulation
Control	STM32G030C8T6, OLED + rotary encoder + 3 keys

BLOCK DIAGRAM



REVISION HISTORY

Rev	MCU	Change
v0.3	CH32V003F4P6	first bench build; NPN Darlington follower
v0.4	CH32V003F4P6	pass device turned round to a PNP
v0.5	CH32V203C8T6	front panel added; the '003 runs out of pins
v0.6	CH32V203C8T6	hardware over-current trip
v1.1	STM32G030C8T6	redrawn as a hierarchy; on-sheet 12 V rail
v1.2	STM32G030C8T6	0–36 V range; analogue nets onto real ADC pins
v1.3	**STM32G030C8T6**	**feedback closed, driver returns tied, sense rescaled**

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Power Path

1 input / PD trigger
2 +12 V gate rail
3 buck-boost power stage
4 gate drivers
5 over-current trip

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Control and UI

7 linear post-regulator
8 current sense
9 MCU
10 user interface
11 programming header

File: control.kicad_sch

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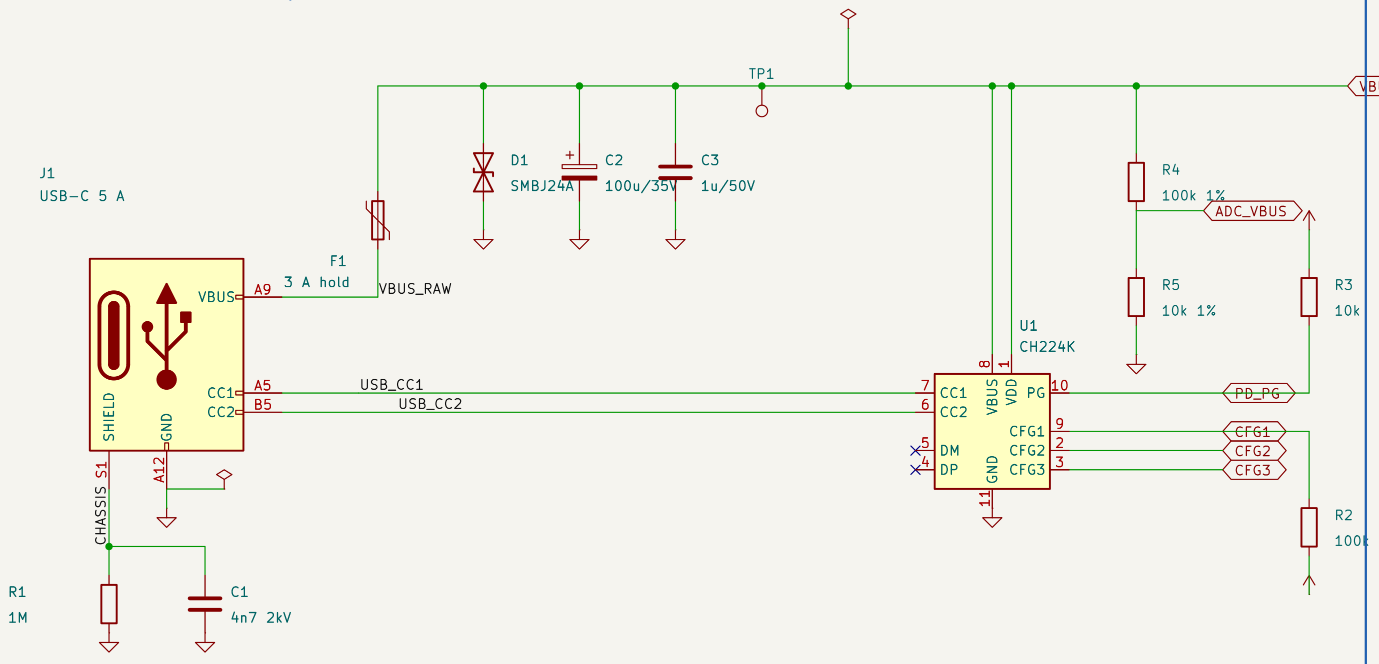
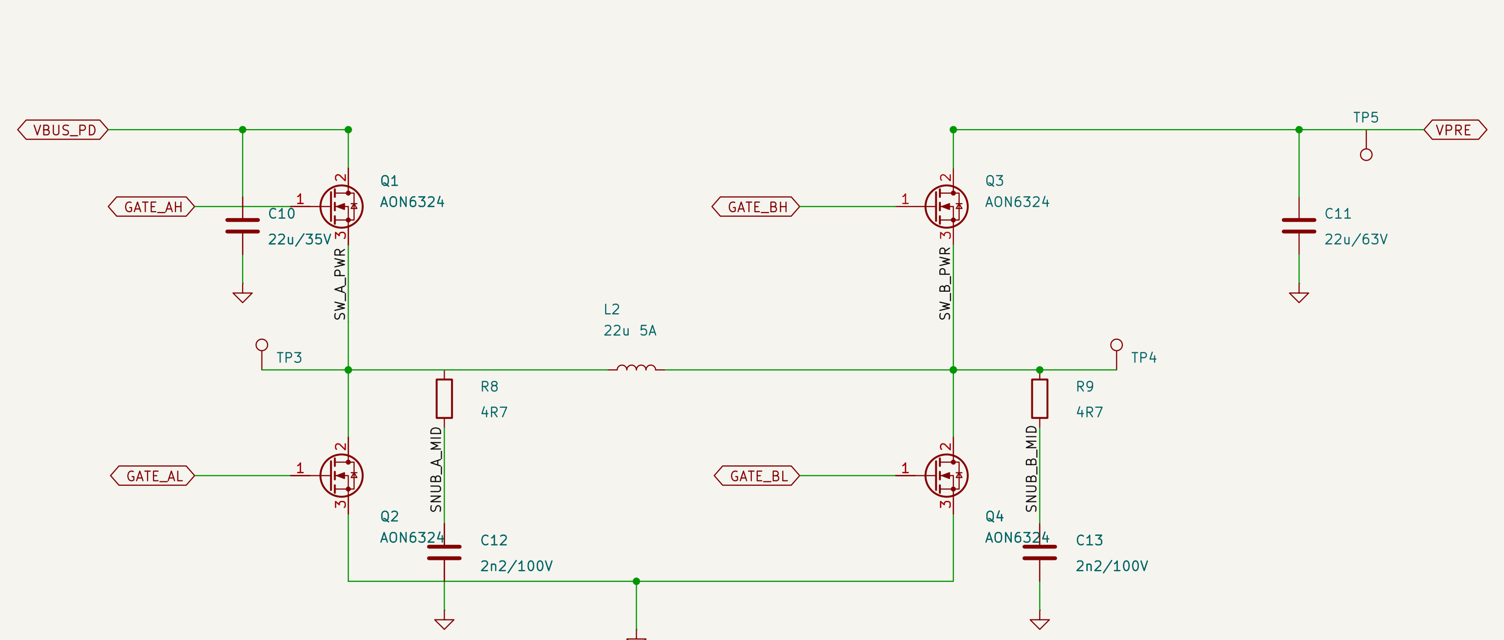
Title page – specification, block diagram, sheet index
Mohammadreza Safaeian – personal project
Sheet: /
File: usbc_pd_psu.kicad_sch
Title: USB-C PD Bench PSU – 4-Switch Buck-Boost + PNP Linear F
Size: A3 | Date: 2026-05-25 | **Rev: v1.3**
KiCad E.D.A. 10.0.3 | Id: 1/3

1 – USB-C INPUT, PROTECTION & PD TRIGGER



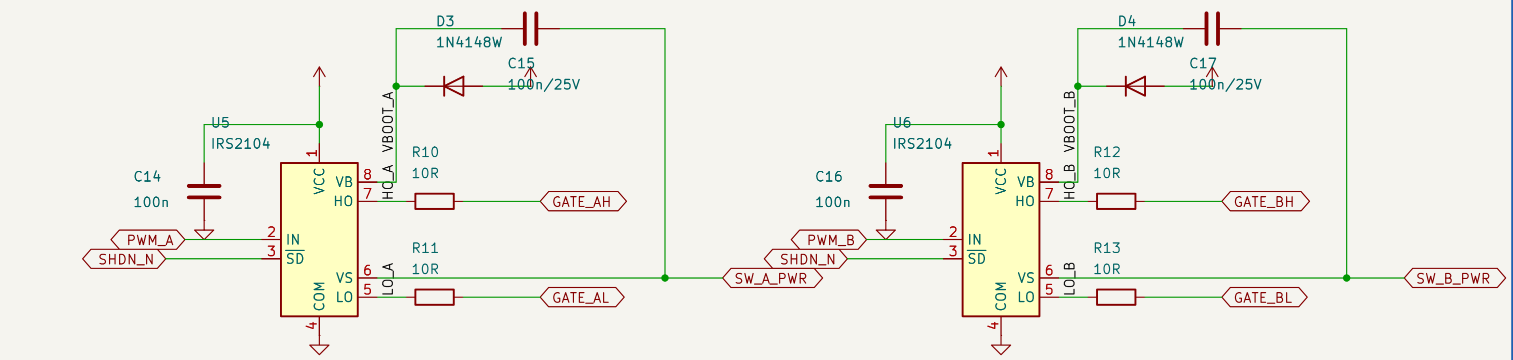
PD contract: CFG1/2/3 select 5 / 9 / 12 / 15 / 20 V. CFG1 is pulled high so a bare board requests 5 V before firmware runs. 500 ms after plug-in the MCU re-reads ADC_VBUS; no rise means a legacy 5 V port and the firmware clamps the output range instead of faulting. The PDO is re-selected as the setpoint changes (table on the title page), with the output disabled during renegotiation.

3 – 4-SWITCH SYNCHRONOUS BUCK-BOOST POWER STAGE



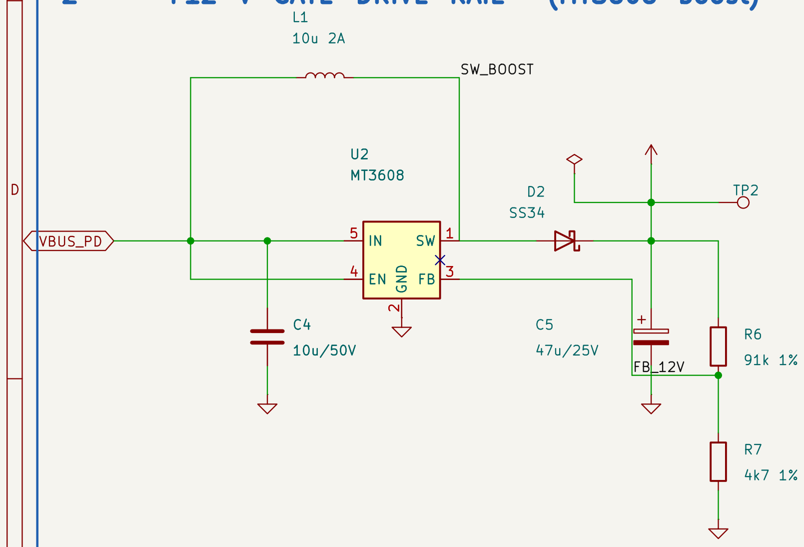
The 4R7 + 2n2 snubbers damp the switch-node ringing set by FET Coss and layout inductance. 60 V FETs, because a 20 V contract plus that ringing will exceed a 30 V part.

4 – HALF-BRIDGE GATE DRIVERS (IRS2104, 520 ns dead-time)



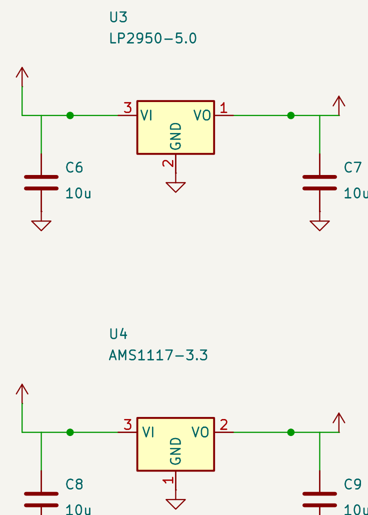
Each bootstrap capacitor returns to its own VS pin, which is tied to the switch node it references – that is what lets the cap refresh and the high-side FET turn on. 10R gate resistors slow the edges enough to keep the ringing in hand.

2 – +12 V GATE-DRIVE RAIL (MT3608 boost)



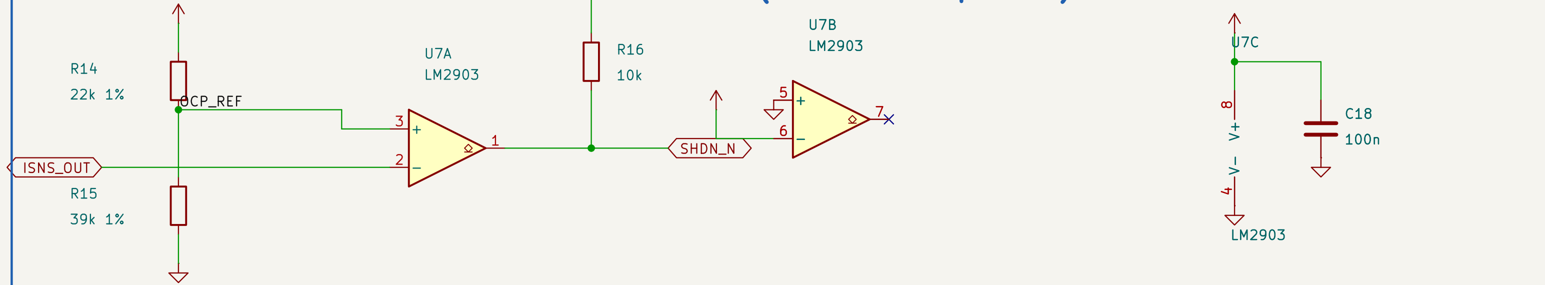
Why an on-board boost: the IRS2104 needs 10–12 V of gate drive (UVLO 8.9 V typ) to put the bridge FETs at their rated RDS(on). This 10 uH / SS34 stage makes 12 V at 150 mA from any contract between 5 V and 20 V, so the drivers behave the same on a legacy 5 V port as on a 20 V one.

6 – LOGIC SUPPLIES



Fed from +12 V, not from VBUS: on a legacy port VBUS sits at ~4.75 V at the connector, below the LP2950 dropout for a 5 V output. Rails therefore come up identically on either port type.

5 – HARDWARE OVER-CURRENT TRIP + LATCH (firmware-independent)



ISNS sits on the inverting input, so an over-current drives the open-collector output LOW and pulls SHDN_N down within ~1 us with no firmware involved. Both IRS2104 SD pins hang on SHDN_N, as does the MCU (open-drain), giving a wired-AND kill.

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Sheet 1 of 2 – input, PD trigger, 12 V rail, buck-boost, drivers, OCP
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Sheet: /Power Path/
File: power.kicad_sch

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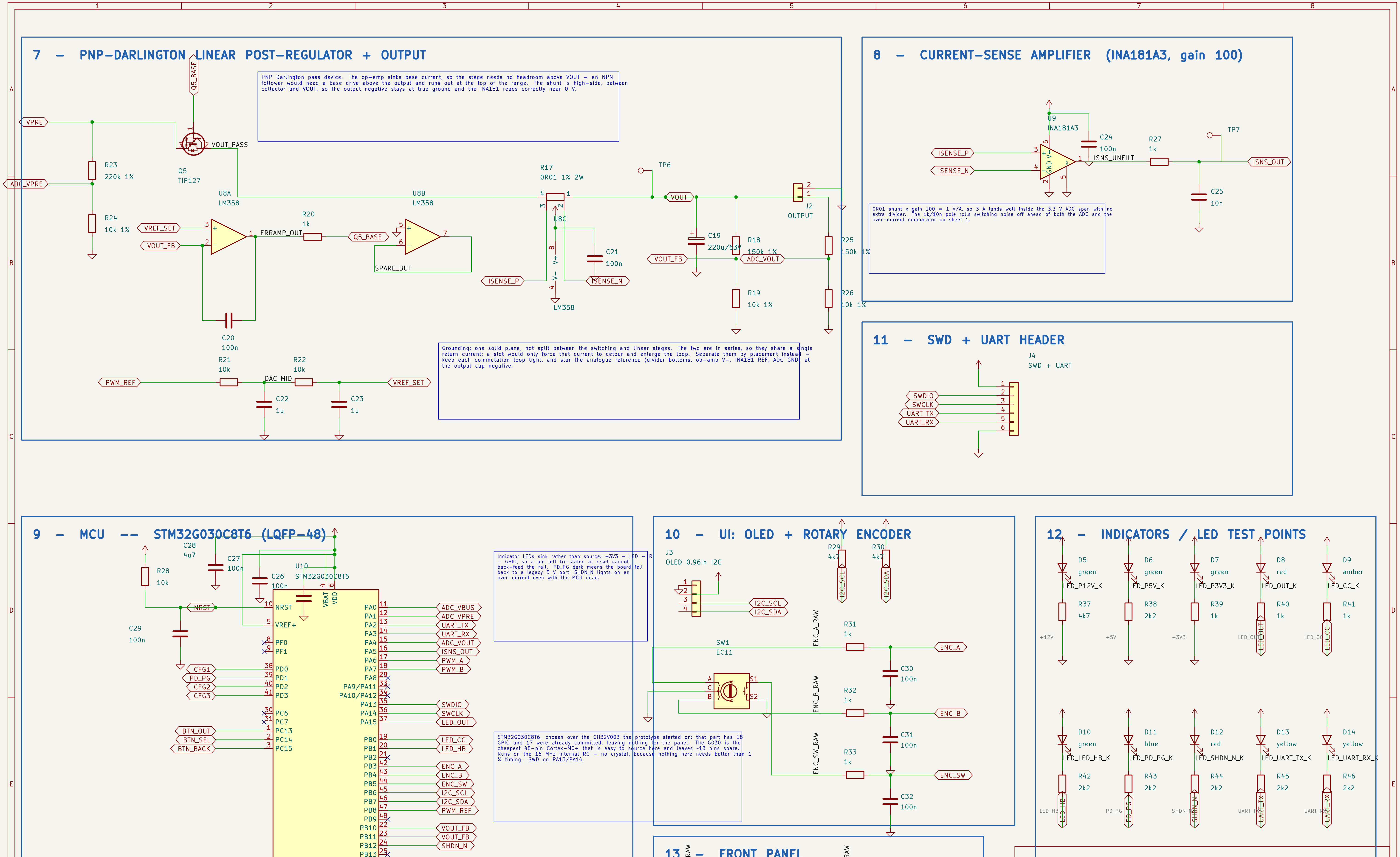

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Sheet 2 of 2 – post–regulator, current sense, MCU, UI, programming
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Sheet: /Control and UI/
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